

From Minimum-Viable-Product (MVP) to Global Climate Resilient App

Climate change significantly impacts traditional agriculture through altered weather patterns, extreme events, and shifting growing seasons, creating an imperative for climate-resilient solutions like hydroponics. While hydroponics offers advantages such as higher yields, improved quality, and reduced water usage, it is not entirely immune to the effects of climate change, including temperature extremes, changes in rainfall, humidity fluctuations, and increased pressure from pests and diseases.

The current Minimum Viable Product Application (aka MVP), serves as a digital logbook which achieves base-level or Level-0 operational goals of Hydroponics farms utilizing Nutrient Film Technique (NFT). However, we would like to expand the architecture as follows:

Level 1: Responsive Operations and Demand Management

Level 2: Balancing farm site selection with crops that can be grown profitably using Climate-Adaptive Intelligence

Level 3: Seed Selection and Crop Rotation that adjusts to micro-climate variance

Details of each level is described below:

Level 1: Responsive Operations and Demand Management

Designing a climate-resilient architecture for a global hydroponics application requires robust informatics and data schemas that can store, process, and leverage environmental metrics and crop tolerance levels. More importantly, the App needs to ensure that each farm meets market demand through real-time monitoring from seeding through to harvest cycles.

The application's FastAPI backend needs schema augmentation in the Supabase-hosted PostgreSQL database to store tenant-specific and climate-relevant configurations. This includes:

- farms table: unit_system (e.g., 'metric', 'imperial'), timezone (e.g., 'Asia/Jerusalem', 'Asia/Singapore'), currency.
- users table: language.
- crops table: Augmented to include ideal_temp_min_celsius, ideal_temp_max_celsius, ideal_light_hours_per_day, nutrient_profile_id, and nft_suitability_notes. This data can be populated from research on heat-tolerant crops.
- nutrient_profiles table: To store target_tds_ppm_min, target_tds_ppm_max, and nutrient_focus.

A core principle is to store data in a single, standard format (e.g., weight in kilograms, temperature in Celsius, volume in liters, dates in ISO 8601) and perform conversions for display purposes at the API layer using libraries like Pint for unit conversions.

Beyond these, the "Hydroponics Climate Resilience Research" highlights the criticality of monitoring and managing several environmental metrics, which should be integrated into the application's schema and logic:

- Water Quality Parameters:

- pH: Optimal range 5.5 to 6.5 for most crops. The HFA should integrate with pH sensors for real-time monitoring, alerts for deviations, and historical data analysis.
- Electrical Conductivity (EC): Reflects total dissolved salts and nutrients. Optimal range varies by crop. The HFA should integrate with EC sensors for continuous monitoring and alerts.
- Dissolved Oxygen (DO): Vital for root development; levels decrease as water temperature increases. The HFA should integrate with DO sensors, provide real-time readings, and alert if levels fall below 5 mg/L. It should also suggest strategies like enhanced aeration.
- Contaminants: Tracking water source information, treatment methods, and quality testing data (e.g., for E. coli, Salmonella, nitrates, heavy metals). Alerts for contamination risks based on external factors like heavy rainfall are crucial.
- Air Temperature and Humidity:
 - Temperature: Generally 18 to 27°C is suitable. The HFA should integrate with temperature sensors, provide real-time monitoring and alerts for deviations.
 - Humidity: Optimal range 50% to 70%. The HFA should integrate with humidity sensors, providing real-time monitoring and alerts.
- Light: The HFA should track natural light levels and allow for control of supplemental lighting, including scheduling and intensity adjustments.

Crucially, the application must allow for customization of these parameters based on specific crop types. A generic set of parameters will not be optimal for all plant varieties. For example, a table outlining optimal ranges for Lettuce, Tomato, Basil, Spinach, and Cucumber, including pH, EC, DO, temperature, and humidity, provides a blueprint for schema design to support this customization.

Level 2: Balancing farm site selection with crops that can be grown profitably using Climate-Adaptive Intelligence

While cities can be significantly impacted by climate change, hydroponics, particularly in the context of urban farming, offers a strategic advantage and a promising solution for food security and sustainability. Hydroponic systems provide a more controlled environment for crop growth, allowing for resource efficiency, especially in water and nutrient usage, by recirculating water and delivering precise nutrients. This controlled environment helps mitigate some adverse climate impacts compared to open-field farming.

Even in urban contexts, which might be among the most impacted by climate change, hydroponics can maximize food production in limited spaces. For instance, in the City of Makati, Metro Manila, Philippines where high temperatures, high humidity, and intense rainfall are common, the application should emphasize features that specifically address these conditions. This includes alerts for high temperatures leading to low dissolved oxygen in the nutrient solution and features for monitoring and controlling humidity to prevent fungal diseases. However, it's also important to acknowledge that hydroponic systems' reliance on a constant power supply makes them vulnerable to power outages, which can be more frequent during extreme weather events.

Level 3: Seed Selection and Crop Rotation that adjusts to micro-climate variance

The impacts of climate change, such as heat waves or sudden temperature drops, can disrupt plant growth and nutrient uptake in hydroponic systems. High air temperatures can also raise the nutrient solution temperature, reducing dissolved oxygen and stressing roots. Therefore, selecting heat-tolerant crop varieties is a crucial first step for climate resilience.

The application's **Crop Suitability Engine** is designed to provide personalized crop recommendations based on a farm's location and climate. This engine compares a location's historical and forecasted monthly average temperature data against a crop's ideal temperature range, drawing from a structured crops table within the application's database.

For regions like Asia, which include target markets such as Singapore, Japan, South Korea, and Malaysia, the selection of heat-tolerant crops becomes particularly important, especially in tropical climates where high temperatures are common. Examples of heat-tolerant crops suitable for NFT (Nutrient Film Technique) hydroponic systems include:

- Cucumber: Ideal temperature range of 22–28°C.
- Bell Pepper: Optimal daytime temperature range of 22–25°C.
- Cherry Tomato: More heat-resilient than larger varieties, with an ideal range of 21–29°C.
- Basil: Ideal temperature range of 26–32°C.
- Mint: Thrives in NFT systems with an ideal range of 25–30°C.
- Okra: Highly heat-tolerant, ideal between 22–35°C.
- Hot Peppers: Suitable and thrive in high temperatures, with an ideal range of 18–30°C.

The application can highlight these crops as "Highly Suitable" or "Conditionally Suitable" based on the farm's climate profile, suggesting necessary environmental controls for the latter.